ControlFace: Harnessing Facial Parametric Control for Face Rigging

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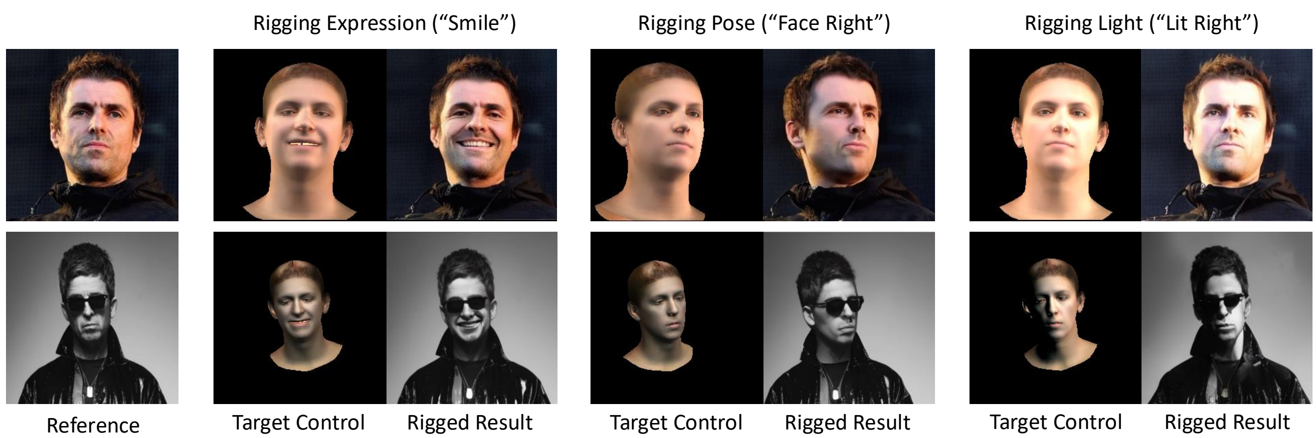

Figure 1. Our **ControlFace** can edit the input face image using explicit facial parametric controls, generating realistic images without compromising the identity and other semantic details such as hairstyle.

Abstract

Manipulation of facial images to meet specific controls such as pose, expression, and lighting, also known as face rigging, is a complex task in computer vision. Existing methods are limited by their reliance on image datasets, which necessitates individual-specific fine-tuning and limits their ability to retain fine-grained identity and semantic details, reducing practical usability. To overcome these limitations, we introduce ControlFace, a novel face rigging method conditioned on 3DMM renderings that enables flexible, high-fidelity control. We employ a dual-branch U-Nets: one, referred to as FaceNet, captures identity and fine details, while the other focuses on generation. To enhance control precision, the control mixer module encodes the correlated features between the target-aligned control and reference-aligned control, and a novel guidance method, reference control guidance, steers the generation process for better control adherence. By training on a facial video dataset, we fully utilize FaceNet's rich representations while ensuring control adherence. Extensive experiments demonstrate ControlFace's superior performance in identity preservation and control precision, highlighting its practicality. Please see the project website: https://cvlab-kaist.github.io/ControlFace/.

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1. Introduction

Manipulating input facial images has long been a fundamental task in computer vision and graphics [2, 26, 43]. Given a face image, the goal of face rigging is to generate an image that corresponds to user controls—lighting or facial pose—without compromising the identity of the face. This task is often complicated by the need to preserve fine details like hairstyle, making it a challenging problem.

Recent advancements in generative models [8, 26, 33, 34, 41] have significantly impacted the development of face rigging techniques. Diffusion models, in particular, have demonstrated remarkable capabilities in producing high-quality, realistic facial images through their iterative refinement processes [17, 41]. These models have shown promise in generating realistic facial images. Leveraging 3D face reconstruction methods [4, 9, 54], enabling the estimation of 3D Morphable Model (3DMM) parameters—such as those of FLAME [24]—directly from an image, many works [7, 10, 19, 25, 28, 42] have utilized 3DMM parameters or renderings as target controls for face rigging. However, existing methods still face several limitations, including difficulties in handling real images [10, 42], the requirement for fine-tuning on images of the same identity [7, 19], and challenges in maintaining fine-grained details from reference images [25, 28]. Addressing these issues is crucial to enhance the practicality and robustness of face rigging approaches in real-world applications.

Notably, conventional diffusion-based face rigging methods [7, 19, 25, 28] rely on single-image-per-individual datasets such as FFHQ [20]. Consequently, these methods are inevitably trained in a reconstruction manner, where only the reference image and its aligned control are available. Specifically, the reference image is embedded through a encoder and the model learns to reconstruct the input reference image with the embedded feature and the aligned control. However, if the feature from the reference image conveys too much information, reconstruction task becomes trivial, which often leads the model to ignore the control signal and simply replicate the reference image making rigging not possible as shown in Fig. 2. Hence, to make the reconstruction non-trivial, they often encode the reference image into a single vector using a network such as pretrained face recognition model [5]. This compression into a single vector, however, may discard essential facial details failing to capture both identity and details simultaneously. This leads to additional individual specific fine-tuning [7, 19]. However, fine-tuning requires images with the same identity as the reference image which may be cumbersome to obtain.

In this paper, we introduce **ControlFace**, a novel face rigging method conditioned on 3DMM renderings, which effectively integrates both target control and reference control extracted from the reference image. Unlike previous methods [7, 10, 25], ControlFace enables flexible edits to lighting, shape, expression, and pose without requiring fine-tuning, while preserving both identity and semantic details of the input reference image. To achieve this, we use dual-branch U-Nets initialized from a large pretrained model: one, termed *FaceNet*, which encodes the reference image to capture its identity and semantic details, and the second *denoising U-Net* that handles the generation. The rich representation encoded by the FaceNet is concatenated with the intermediate feature of the denoising U-Net and integrated through self-attention layer. For better control adherence we introduce the control mixer module (CMM), which embeds the correlated features between the target and reference controls. Additionally, we introduce a reference control guidance (RCG) mechanism that uses the reference control as a null condition label to improve grounding, inspired by Classifier-Free Guidance (CFG) [16]. ControlFace is trained on a facial video dataset [52], which provides paired reference images and target images aligned with the desired control. This training approach allows us to avoid reconstruction-based training and hence fully leverage FaceNet's rich generative representation without ignoring the target control.

We conduct extensive experiments and ablation studies to rigorously evaluate the effectiveness of our proposed method. We demonstrate ControlFace's superior performance in identity preservation, semantic consistency, and control precision compared to existing methods, by evaluating on multiple unseen faces [20] We also show that ControlFace outperforms fine-tuning-based methods [7] without the need for the cumbersome additional training. Ablation studies further highlight the distinct contributions of all the components in our model. These results validate our approach and illustrate its potential for real-world applications in face rigging.

Figure 2. **Limitations of reconstruction-based training.** We compare the results of our model trained on an image dataset [20] in a reconstruction setup and on a video dataset [52] with paired samples created by randomly selecting two frames from each video. The results by reconstruction-based training often ignores the target control at inference.

2. Related Work

Generative Face Editing. Since the introduction of GANs [11], face generation has received significant attention. Early GAN-based methods [13, 39, 44] leveraged the disentangled latent space of StyleGAN [20] to manipulate facial attributes by finding specific directions in latent space. Other approaches [6, 10, 26, 38, 42] incorporated 3DMMs [2, 3] to conditionally control pose and expression through 3D structural guidance, improving realism and consistency.

Recently, diffusion models have emerged as a powerful alternative for generative face editing due to their stable training and high-quality output. For identity-preserving generation and manipulation, recent diffusion-based works either incorporate facial features directly [12, 25, 45, 46, 48], train networks to encode images [7, 19, 30, 49], or leverage pretrained diffusion U-Nets for rich representations [18, 47, 53].

Controllable Generation for Diffusion Models. Methods for conditioning diffusion models with spatial controls initially relied on concatenation [34–36]. The advent of large-scale pretrained text-to-image models [8, 34] has leveraged sophisticated text embeddings [31, 32] processed through advanced attention mechanisms, driving research into finer control over the generation process [23, 27, 29, 50]. Recently, ControlNet [50] has enabled precise control over attributes like depth, edges, and poses by integrating timestep-specific control features into the diffusion U-Net. In face editing, approaches [7, 12, 19, 25] have adopted similar architectures with 3DMMs to allow explicit facial control. However, these methods often struggle to encode fine details from the reference image. To overcome this limitation, we employ dual-branch structure: one for encoding the reference facial image and the other for generating manipulated image.

3. Preliminaries

3D Morphable Face Models (3DMMs). 3DMMs [2, 3, 24] are facial parametric models designed to accurately represent facial pose, shape, and expression using a compact latent. FLAME [24] which is one of the widely used 3DMM, takes shape, pose, and expression as inputs and outputs a face mesh. In this work, we leverage DECA [9] model which is capable of predicting the FLAME [24] parameters together with the orthographic camera parameters, albedo, and lighting in spherical harmonics from a facial image. With these parameters we can apply Lambertian reflectance to acquire surface normals, albedo, and Lambertian rendering. These renderings offer pixel-aligned control, providing both geometric and texture information that suitable for face rigging. Hence, we leverage these renderings as our control.

Latent Diffusion Models (LDM). LDM [34] is a generative model that progressively learns to denoise random Gaussian noise in the latent space of VAE [21]. During training, an input image X is encoded into a latent representation z via a VAE encoder. This latent z is then perturbed with Gaussian noise ϵ through a predefined diffusion forward process at a randomly selected timestep t, producing the noisy latent z_t. The model's core objective is to predict ϵ, effectively learning to reverse the noise. The training objective can be written as:

$$\mathcal{L}_{LDM} = \mathbb{E}_{z,t,\epsilon\sim\mathcal{N}(0,1)}[\|\epsilon_\theta(z_t;t) - \epsilon\|_2^2]. \quad (1)$$

Classifier-Free Guidance (CFG). To enhance conditional generation of diffusion model during inference, CFG [16] reformulates the predicted noise steering the generation process away from the unconditional distribution and towards the conditional distribution. This allows us to sample from a sharper conditional distribution, enhancing the condition fidelity. The modified prediction can be formulated as follows:

$$\hat{\epsilon}_\theta(\cdot, C) = \epsilon_\theta(\cdot, \varnothing) + w\left(\epsilon_\theta(\cdot, C) - \epsilon_\theta(\cdot, \varnothing)\right), \quad (2)$$

where $\varnothing$ denotes an unconditional label such as empty text, C denotes the conditional input and w is a guidance scale. Hence, $\epsilon_\theta(\cdot, \varnothing)$ is the unconditional prediction and $\epsilon_\theta(\cdot, C)$ is the conditional prediction. Many diffusion models, including text-to-image models [1, 8, 33, 34], employ this technique to achieve high-quality generation.

4. Method

4.1. Overview

Given a facial reference image $X_R \in \mathbb{R}^{H\times W\times 3}$ with height H and weight W, our objective is to generate a target facial image X_T that retains the identity and detailed attributes of X_R, while aligning with an image-aligned target control $D_T \in \mathbb{R}^{H\times W\times 9}$, which is defined by 3DMM renderings [9, 24] consisting of surface normal, albedo, and Lambertian renderings.

We employ dual branch U-Nets: the proposed FaceNet to capture the local and fine-grained appearance of X_R and a denoising U-Net for generation. FaceNet's detailed features are injected into the denoising U-Net through the augmented self-attention layer. Additionally, we utilize CLIP [31] image encoder to provide coarse visual features to both U-Nets. To guide manipulation by control signal, we present a lightweight encoder called face controller, which embedds D_T directly into the denoising U-Net. However, as it only encodes D_T, the model cannot fully map the relation between X_T and X_R, limiting control adherence. Hence, we propose control mixer module (CMM) with multiple cross-attention layers to embed correlated features between D_T and D_R which is then fused into the augmented self-attention layer to guide the model's attention effectively. D_R is another 3DMM renderings extracted from the reference image. Fig. 3 illustrates our overall architecture. For further control adherence, we employ reference control guidance (RCG) during inference. Each component is explained in details in the following sections.

4.2. Acquiring Paired Quadruplets for Face Rigging

As shown in Fig. 2, training our model with image datasets where $X_R = X_T$ like previous methods [7, 19] may lead the model to replicate X_R and ignore D_T. Hence, we leverage facial video dataset, CelebV-HQ [52], which contains 35,666 facial video clips involving 15,653 identities with various actions. From each video, we randomly select two frames, assigning one as X_R and the other as X_T. We then extract image-aligned surface normals, albedo, and Lambertian rendering from each frames. The extracted renderings from the reference image serve as the reference control D_R, while those from the target image serve as the target control D_T. Consequently, our dataset consists of a set of quadruplets consisting of $\{X_R, X_T, D_R, D_T\}$.

4.3. Injecting Appearance of Reference Image

To acquire strong image representations that preserve identity and fine details during generation, the reference image

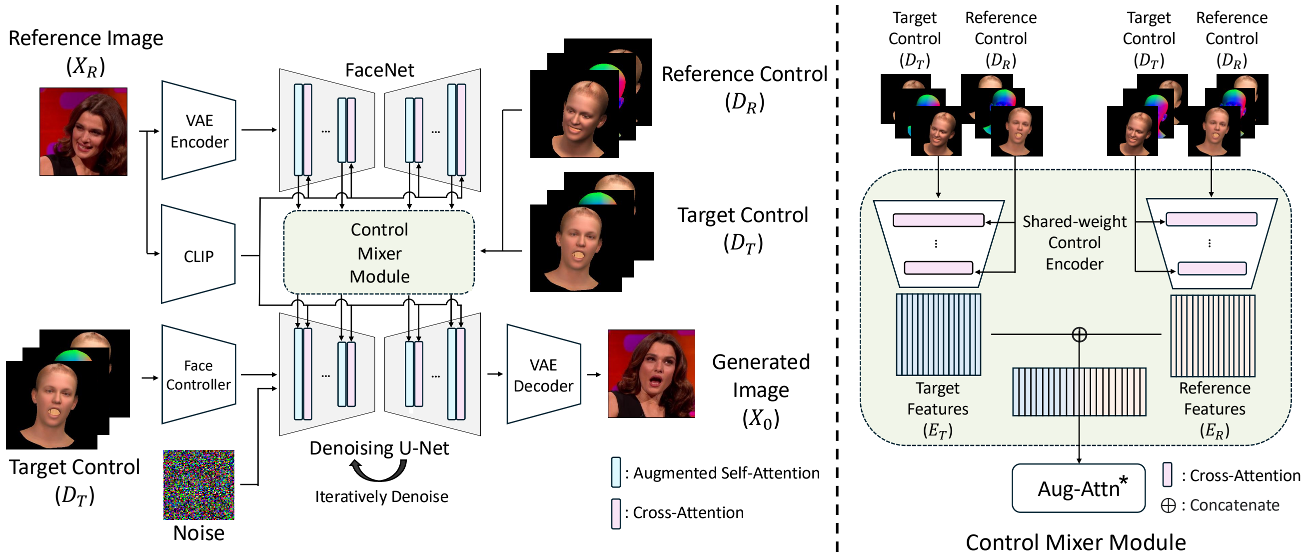

Figure 3. **Overall Architecture.** ControlFace encodes the reference image X_R into the FaceNet and CLIP image encoder for identity and semantic preservation. For face control, the target control D_T is incorporated into the denoising U-Net through face controller. To enhance the control adherence, the correlated feature between reference control D_R and target control D_T is acquired from the proposed control mixer module.

is encoded using both the CLIP [31] image encoder and the proposed FaceNet. The CLIP image embeddings, which capture high-level semantic information, are integrated into both denoising U-Net and FaceNet through cross-attention layers. Simultaneously, FaceNet encodes the reference facial image to extract detailed features which are then incorporated into the denoising U-Net through augmented self-attention mechanism [18, 37, 40, 47]. Since FaceNet and the denoising U-Net share the same architecture, they integrate seamlessly in self-attention by concatenating keys and values from both U-Nets and applying attention operation with queries from the denoising U-Net. Let Q, K, $V \in \mathbb{R}^{lxd}$ denote query, key, value from the denoising U-Net where l and d denote length and dimensionality. Also, let K^{face}, $V^{\text{face}} \in \mathbb{R}^{lxd}$ represent key, and value from the FaceNet, respectively. Then the augmented self-attention Aug-Attn is formulated as follows:

$$\text{Aug-Attn} = \text{Softmax}(\frac{Q[K, K^{\text{face}}]^T}{\sqrt{d}})[V, V^{\text{face}}], \quad (3)$$

where $[\cdot, \cdot]$ denotes concatenation, and $\text{Softmax}(\cdot)$ means a softmax operation.

4.4. Incorporating Target Controls

Face Controller. To manipulate the reference image, we directly encode the target control D_T using a CNN-based shallow network which we denote face controller. The encoded features from the face controller are added to the first layer of the denoising U-Net. We have tested alternative diffusion conditioning methods, like ControlNet [50] and ControlNeXt [29], but led to worse performance with increased computational complexity. The ablation study on the conditioning methods can be found in the Section 5.4.

Control Mixer Module. CMM outputs correlated embeddings for both controls D_R and D_T, denoted E_R, and E_T, respectively. CMM encodes these controls using two shared-weight control encoders. The control encoder consists of multiple convolution layers along with the cross-attention layers which is capable of modeling the interaction between the two control effectively. Specifically, when one control is input, the other serves as a conditioning input, passed through a convolution layer to match the resolution with the input.

The structure of CMM is illustrated in Fig. 3. The resulting embeddings E_R and E_T are in four different resolutions corresponding to the four different resolutions of U-Net activation maps. These embeddings are fused with features from the two U-Nets in the augmented self-attention layer. We modify Eq. 3 by adding E_R and E_T to the corresponding keys and values and denote it as Aug-Attn^*. This can provide guidance on where the model should attend to. The modified augmented self-attention can be formulated as:

$$\text{Aug-Attn}^* =$$
$$\text{Softmax}(\frac{(Q + E_T)[K + E_T, K^{\text{face}} + E_R]^T}{\sqrt{d}})[V, V^{\text{face}}]. \quad (4)$$

4.5. Training Objective

Since we perform the generation process in the latent space of a VAE [21], we encode X_R and X_T via the VAE encoder

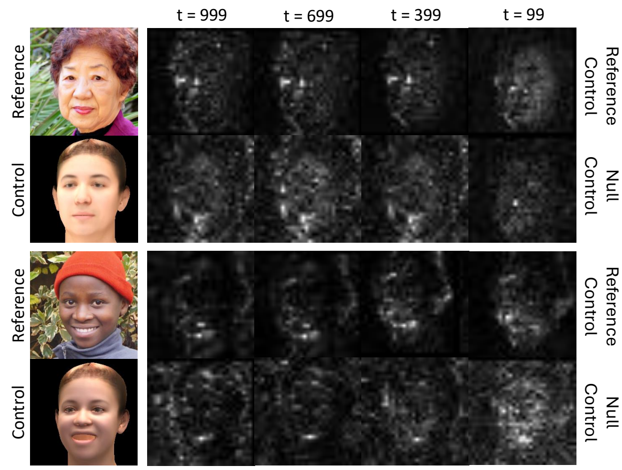

Figure 4. **Visualization of Reference Control Guidance.** We visualize the deltas, $\epsilon_\theta(\cdot, D_T) - \epsilon_\theta(\cdot, \varnothing)$ and $\epsilon_\theta(\cdot, D_T) - \epsilon_\theta(\cdot, D_R)$, which corresponds to CFG [16] applied to face controller input and RCG, respectively, across different timesteps t. The first and third row display RCG deltas whereas second and fourth row show the CFG deltas. The former shows noisy deltas over all the timesteps.

and acquire latents z_R and z_T, respectively. Our model $\epsilon_\theta(\cdot)$ takes z_T,z_R, D_T, and D_R as input and learns to predict random Gaussian Noise ϵ which is used to perturb z_R. Similar to Eq. 1, our training objective is as follows:

$$\mathcal{L} = \mathbb{E}_{z,t,\epsilon\sim\mathcal{N}(0,1)}[||\epsilon_\theta(z_{T,t}; t, z_R, D_T, D_R) - \epsilon||_2^2], \quad (5)$$

where $z_{R,t}$ is the perturbed z_R with predefined diffusion forward schedule with randomly selected timestep t.

4.6. Reference Control Guidance

At inference, we further enhance the control adherence with a novel guidance technique. To achieve this, we first naively apply CFG to face controller input. This can be formulated as follows:

$$\hat{\epsilon}_\theta(\cdot, D_T) = \epsilon_\theta(\cdot, \varnothing) + w\big(\epsilon_\theta(\cdot, D_T) - \epsilon_\theta(\cdot, \varnothing)\big), \quad (6)$$

where $\varnothing$ is a null input for the face controller, similar to Eq. 2. However, Tab. 5 shows that the improvement is negligible compared to the result without any guidance. We suspect that this is because the null condition cannot provide a good grounding for where to deviate away from. Therefore, our proposed guidance method, RCG, utilizes D_R instead of null conditioning for better grounding. We define it as follows:

$$\hat{\epsilon}_\theta(\cdot, D_T) = \epsilon_\theta(\cdot, D_R) + w\big(\epsilon_\theta(\cdot, D_T) - \epsilon_\theta(\cdot, D_R)\big). \quad (7)$$

In Fig. 4, we visualize each difference, $\epsilon_\theta(\cdot, D_T) - \epsilon_\theta(\cdot, \varnothing)$ and $\epsilon_\theta(\cdot, D_T) - \epsilon_\theta(\cdot, D_R)$, on various timesteps. Although the difference of both guidances tends to focus on facial area on lower timesteps, the former is noisy over all timesteps which can compromise the generation process. On the other hand, the difference of RCG shows relatively clean face-aligned estimates over all timesteps. Hence, we leverage Eq. 7 for generating our output X_0.

5. Experiments

5.1. Implementation Details

We train our model on CelebV-HQ [52] dataset, resizing all frames to 256×256 pixels. We extract 3DMM renderings also of shape 256×256 from each frame leveraging DECA [9]. The weights of the FaceNet and the denoising U-Net are first initialized from a variant of pretrained Stable Diffusion v1.5 [34], which is a large text-to-image model fine-tuned to accept CLIP [31] image embeddings as input. We present more implementation details in the Appendix **??**.

5.2. Comparison

Baselines. We compare our method with four publicly available 3DMM-controllable face generation methods: GIF [10], CapHuman [25], Arc2Face [28], and DiffusionRig [7]. GIF is a 3DMM-conditional GAN-based approach utilizing the StyleGAN [20] generator, but it does not take a reference image as input. Other methods are diffusion-based. CapHuman and Arc2Face utilize a face recognition model [5] feature for identity preservation, whereas DiffusionRig employs ResNet [14] to encode the reference image features. CapHuman also accepts text input for additional control; therefore, we fix the input text to "a photo of a person" in our experiments to ensure a fair comparison.

We evaluate ControlFace and the baselines on the FFHQ [20] dataset, which consists of over $70,000$ high-quality facial images. Notably, all baselines except CapHuman were trained on FFHQ, whereas our method was not. Additionally, ControlFace is trained on CelebV-HQ [52], which contains approximately $15,000$ unique identities—fewer than FFHQ. This demonstrates our method's robustness and generalizable capability.

Qualitative Comparison. We compare our generated results with the baselines on three identities from the FFHQ dataset. Fig. 5 shows the results. We present rigged outputs with variations in three attributes—pose, expression, and lighting—for all methods except Arc2Face, as it does not support lighting control. For visualization, we display only the Lambertian rendering of the control.

While all methods produce results aligned with the control parameters, GIF fails to preserve identity as it does not incorporate any image-related features. CapHuman and Arc2Face maintain the identity but lose semantic details like hairstyle and background since they utilize only facial features. DiffusionRig shows good performance in identity and detail preservation; however, the identity tends to change

Figure 5. **Qualitative Results.** We compare the results of rigging pose, expression, and light with four different baselines [7, 10, 25, 28]. The reference images are from the FFHQ [20] dataset. Compared to the baselines ControlFace aligns with the target control, maintaining the identity and details in the reference image.

when the control differ significantly from those of the reference image. Compared to all baselines, our method exhibits superior performance in both identity and detail preservation while adhering to the target controls. We present more generated results on FFHQ dataset in the Appendix D.1.

Moreover, since DiffusionRig originally requires additional training on a personal album for optimal performance, we fine-tune it on the reference image. As shown in Fig. 6, while fine-tuning improves DiffusionRig's identity and detail preservation, it still struggles with identity shifts under large pose changes. In contrast, our method maintains consistent identity even under large pose variations, demonstrating robustness and fidelity in generated results.

Quantitative Comparison. We measure the DECA re-inference error using the same settings as previous works [7, 10, 19] to evaluate fidelity to the target controls. Specifically, we generate 1,024 images by varying shape, lighting, expression, and pose parameters. For each generated image, we use DECA to re-inference the 3DMM parameters. We then calculate the Root Mean Square Error (RMSE) between the FLAME mesh rendered with original control parameters and the newly inferred ones. For lighting, we compute the RMSE directly on the spherical harmonics coefficients. Tab. 1 shows that our method achieves the best control adherence on average.

We evaluate image quality using three metrics: identity preservation (ID) score via cosine similarity of features from a face recognition model [5], Fréchet Inception Distance (FID) [15] for image quality, and Learned Perceptual Image Patch Similarity (LPIPS) [51] for semantic consistency. These metrics are computed on the same set of 4096 images used for DECA re-inference error evaluation. ID and LPIPS are not calculated for GIF, as it does not use

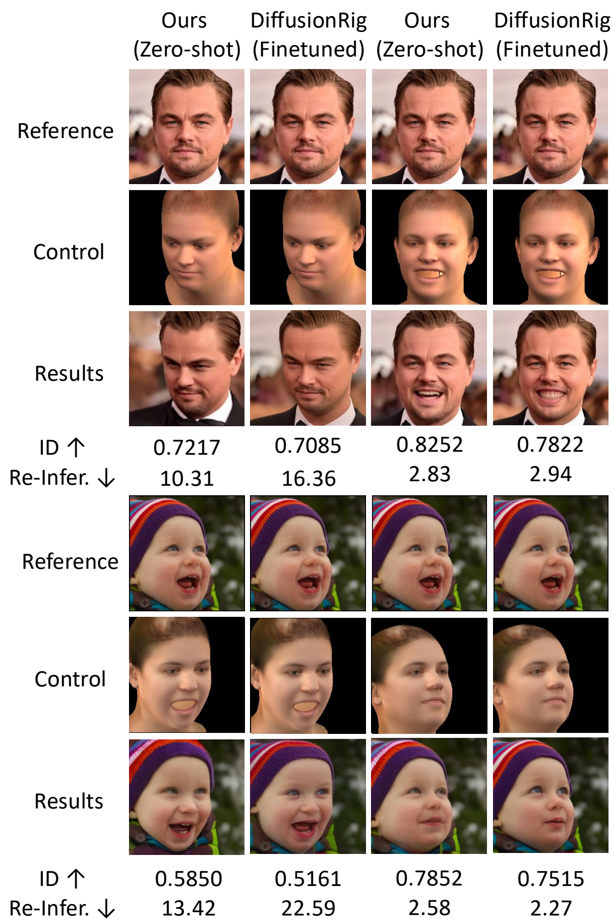

Figure 6. **Comparison with Fine-tuning Method.** We compare the results with the fine-tuned DiffusionRig [7] on two reference images. We show the generated results along with the identity similarity (ID) [5] and DECA [9] re-inference error.

a reference image. Results in Tab. 2 show our method achieves the lowest FID and LPIPS, indicating high image quality and semantic consistency. While Arc2Face scores highest in identity preservation due to explicit facial feature conditioning, its high LPIPS score indicates a lack of background preservation. In contrast, our method not only preserves identity but also maintains semantic consistency with the reference image, offering a balanced performance across all evaluated metrics. Also, for a fair comparison, we train DiffusionRig on CelebV-HQ and apply the same evaluation metrics which is illustrated in the Appendix C.1.

We conduct a user study for additional evaluation, following ImagenHub standards [22]. Participants rate each model on semantic consistency (SC), measuring alignment between the condition and the generated image, and perceptual quality (PQ), assessing image quality. We compare our model with CapHuman and DiffusionRig, both of which use reference images and lighting as conditions. Tab.3 presents

Table 1. **Quantitative Results on Control Adherence.** We compare the DECA [9] re-inference errors following previous work [10] with four different baselines [7, 10, 25, 28]. Our method, ControlFace, shows the best control adherence on average.

	Light↓	Shape ↓	Exp. ↓	Pose ↓	Avg. ↓
GIF [10]	17.04	2.29	8.16	8.17	8.91
CapHuman [25]	15.16	2.65	6.68	19.03	10.40
Arc2Face [5]	-	3.13	13.83	15.37	10.77
DiffusionRig [7]	6.31	**2.11**	5.58	**6.26**	5.06
ControlFace (Ours)	**3.75**	2.56	**5.43**	7.67	**4.85**

Table 2. **Quantitative Results on Identity Preservation and Image Quality.** We evaluate identity similarity (ID) [5] and LPIPS [51] to assess appearance preservation of the reference image, and FID [15] to measure overall image quality.

	ID↑	FID ↓	LPIPS ↓
GIF [10]	-	24.48	-
CapHuman [25]	0.4356	59.78	0.4556
Arc2Face [5]	**0.7825**	17.82	0.5253
DiffusionRig [7]	0.2042	23.05	0.3758
ControlFace (Ours)	0.7586	**15.50**	**0.1429**

Table 3. **User Study.** We ask the participants to evaluate the model on condition alignment (SC) and perceptual quality (PQ). Our model achieves the highest score on all metrics.

	SC↑	PQ ↑	OV ↑
CapHuman [25]	0.4086	0.7019	0.4545
DiffusionRig [7]	0.3533	0.5072	0.3473
ControlFace (Ours)	**0.8750**	**0.8605**	**0.8486**

SC, PQ, and the overall score (OV), which is the geometric mean of SC and PQ demonstrating the final score of the model. Our model achieves the highest scores across all metrics. Further details of the user study is available in the Appendix B.

5.3. Result on Out-of-Domain images

To further evaluate the zero-shot generalization capability of our model, we test ControlFace on out-of-domain reference images that significantly differ from the real face data leveraged during training. Specifically, we employ animation-like facial images as reference images. As shown in Fig. 7, our model successfully generates rigged results for these animation faces, demonstrating that it can effectively handle outside the scope of its training data. We present more examples in the Appendix D.2.

5.4. Ablation Studies

Scaling Reference Control Guidance. We examine the effect of guidance scales w on control adherence. From here, CFG* and CFG† refer to CFG applied to the face controller and CMM, respectively, both incorporating null inputs. On the other hand, CFG represents applying a null

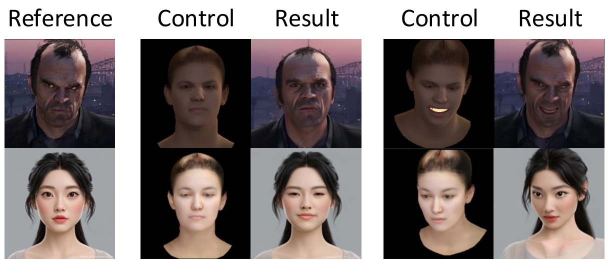

Figure 7. **Results on Out-of-domain Data.** We test ControlFace on animation-like faces to illustrate the robustness and generalizability.

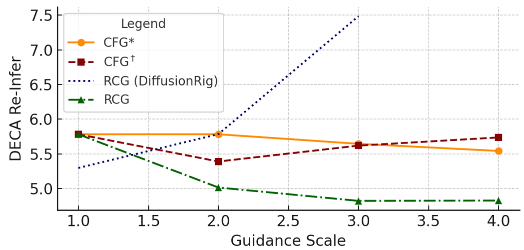

Figure 8. **Ablation on Guidance Scale.** We plot the DECA [9] re-inference error for guidance scales w from 1.0 to 4.0. Increasing w improves errors for RCG, CFG*, and CFG† to some extents, but greatly worsens for DiffusionRig [7].

input to the CLIP encoder, following the approach used in most text-to-image models [8, 33, 34]. Since RCG requires no null condition training, we also apply it to DiffusionRig, evaluating on 128 randomly selected FFHQ reference images. Fig. 8 shows that increasing the scale improves DECA re-inference error for RCG, CFG*, and CFG† to varying extents, but significantly reduces performance for DiffusionRig. We suspect this is due to DiffusionRig's struggle to maintain identity consistency under different controls, causing conflicting identities and poor alignment when using both target and reference controls.

Ablation on Different Conditioning Modules. We attach different kinds of widely used conditioning modules [29, 50] to see their effects on control adherence. On top of FaceNet archetecture, without CMM, we attach various conditioning modules, applying CFG during inference. Tab. 4 compare face controller with ControlNet [50] and ControlNeXt [29] showing the number of parameters and DECA re-inference error. Face controller performs the best while having the least number of trainable parameters.

Ablation on Architecture. Starting with FaceNet and Face Controller, we sequentially add CMM, CFG [16], CFG*, and our proposed RCG, with results in Tab. 5. Incorporating CMM improves alignment and image quality but slightly reduces identity similarity compared to FaceNet. This occurs because enhanced control adherence requires adjustments to attributes like expression, pose, or lighting, which may inadvertently alter identity-related features.

Table 4. **Conditioning Module Ablation.** We plot parameter counts and DECA [9] re-inference errors for various conditioning modules.

	# of params.	Re-Infer.
ControlNet [50]	∼360M	8.37
ControlNeXt [29]	∼3M	8.06
Face controller (Ours)	**∼1M**	**7.87**

Table 5. **Architecture Ablation.** We plot DECA [9] re-inference error, identity similarity score [5], FID [15], and LPIPS [51] for each architecture.

	Component	Re-Infer.↓	ID ↑	FID ↓	LPIPS
(I)	FaceNet	7.13	**0.8234**	32.45	**0.1321**
(II)	(I) + CMM	5.78	0.7520	**15.35**	0.1384
(III)	(II) + CFG	5.77	0.7586	15.50	0.1429
(IV)	(II) + CFG*	5.75	0.7308	15.43	0.1400
(V)	**(II) + RCG (Ours)**	**4.85**	0.7586	15.50	0.1429

Consequently, FaceNet, with poorer alignment, achieves the highest identity preservation. Our proposed RCG method compensates for the marginal decrease in ID score by achieving better control adherence and image quality than FaceNet. Also, RCG exhibits better control alignment among other guidance methods without sacrificing the identity. We additionally provide ablation study on different inputs and architecture for CCM in the Appendix C.3.

5.5. Limitation

ControlFace relies on DECA [9] for acquiring 3DMM renderings which may not capture the finer facial details needed for optimal performance. Adopting a more advanced 3D face estimation model [4, 54] could enhance the model's overall performance. Additionally, ControlFace is currently trained on a single facial video dataset [52], limiting its exposure to a variety conditions and identities. Incorporating additional video datasets, as well as multiview facial datasets, could further improve the model's generalizability.

6. Conclusion

In this paper, we introduce ControlFace, a face rigging method that addresses limitations arising from reliance on image datasets, such as loss of fine details and need for fine-tuning. Our dual-branch U-Net is capable of capturing the identity and intricate details of the reference image. To improve control adherence, we leverage CMM which learns embedding the relationship between the target and reference controls, and RCG which utilizes reference control for better grounding. Training on a facial video dataset improved identity and detail preservation. Experiments and ablations confirm ControlFace's superior performance in identity consistency, control adherence, and image quality, showing its practicality for real-world applications.

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